



Self-supervised Learning for Electroencephalogram: A Systematic Survey

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Electroencephalography (EEG) is a non-invasive technique to record bioelectrical signals. Integrating supervised deep learning techniques with EEG signals has recently facilitated automatic analysis across diverse EEG-based tasks. However, the label issues of EEG signals have constrained the development of EEG-based deep models. Obtaining EEG annotations is difficult and requires domain experts to guide collection and labeling, and the variability of EEG signals among different subjects causes significant label shifts. To solve the above challenges, self-supervised learning (SSL) has been proposed to extract representations from unlabeled samples through well-designed pretext tasks. This article concentrates on integrating SSL frameworks with temporal EEG signals to achieve efficient representations and proposes a systematic survey of the SSL for

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EEG signals. In this article, (1) We introduce the concept and theory of self-supervised learning and typical SSL frameworks. (2) We provide a comprehensive survey of SSL for EEG analysis, including taxonomy, methodology, and technical details of the existing EEG-based SSL frameworks, and discuss the differences between these methods. (3) We investigate the adaptation of the SSL approach to various downstream tasks, including the task description and related benchmark datasets, and further explore its application in large-scale pre-trained foundation models for EEG signals. (4) Finally, we discuss the potential directions for future SSL-EEG research.

CCS Concepts: • **Theory of computation** → **Models of computation**; • **Applied computing** → **Computational biology**; • **Computing methodologies** → **Artificial intelligence**;

Additional Key Words and Phrases: Self-supervised learning, electroencephalogram, contrastive learning, representation learning

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1 Introduction

Electroencephalography (EEG) is a neurophysiological technique that records and measures the brain's electrical activity. The EEG signals are collected in a non-invasive way that involves placing electrodes on the scalp to measure and record the electrical impulses generated by the brain [133]. EEG signals, as external representations of internal brain neural activity, encapsulate rich neural information associated with various brain stimuli. This characteristic has made EEG a widely studied modality for addressing diverse real-world tasks, such as epilepsy detection [47], emotion recognition [14], sleep research [116], and applications in brain-computer interfaces [4]. Therefore, the EEG signal is an incredible tool in neuroscience and possesses an exceptionally high clinical utility, which generally became the research focus of physiological signals.

Recently, with the fast development of deep learning and artificial intelligence, machine learning and deep learning models have been integrated with labeled samples to complete different classification [127], regression [117], and generation [41] tasks. The combination of intelligent algorithms and labeled EEG datasets with supervised learning modes has emerged as a powerful tool to enhance the analysis and interpretation of EEG data. Traditional machine learning methods such as **Support Vector Machine (SVM)** [108], **Random Forest** [110], and **Multi-Layer Perceptron (MLP)** [114] have demonstrated their efficiency in detecting significant patterns from different hand-crafted EEG features [63]. Some simple EEG-based tasks, such as EEG-based event classification, emotion recognition, epilepsy detection, and motor imagery classification, can be automatically performed by machine learning models [37, 103, 165]. The end-to-end deep learning frameworks composed of the **Convolutional Neural Network (CNN)** [101], **Long-Short Term Memory Network (LSTM)** [157], **Transformer**[136], and other networks are implemented to model the spatial correlation between electrodes and the temporal variation of the EEG signals [5, 93, 131]. Deep learning methods contain more parameters and complex network structures, with stronger learning and expression abilities to extract physiological information and recognize complex patterns [44, 142]. Adequate labeled EEG data and powerful deep learning models are critical elements for intelligent EEG analysis [83]. Moreover, relying on the large amount of high-quality labeled EEG data, deep models trained with supervised modes can accomplish complex EEG tasks [1, 35, 107].

The most critical challenge intelligent EEG analysis faces is the scarcity of labeled samples. Training deep models demands extensive labeled data. However, obtaining a large-scale labeled EEG signal for model training is impractical. The annotation of the EEG signal necessitates manual intervention from experts well-versed in neurophysiology who possess profound familiarity with the distinctive features of interest embedded within the EEG data. The high costs and the need for expert knowledge in the annotation process make constructing EEG datasets extremely challenging. In addition, the scarcity of specific brain states significantly affects the acquisition of EEG signals [112]. For example, abnormal emotion states and seizure states are relatively rare among the subjects, making it more difficult for sample collection. Therefore, building annotated EEG datasets for training deep models is constrained by various factors. It necessitates the involvement of domain experts [45], demanding substantial time and cost [25], which poses a significant challenge for the application of supervised learning in EEG analysis.

Besides, supervised EEG analysis faces the inconsistency problem [67, 169], which severely impacts the effectiveness of supervised learning. Interpreting EEG signals often involves subjectivity and variability among the subjects and evaluators [29]. First, some tasks generally collected signals annotated by the participants, which have a vital subjective component and do not necessarily represent the actual states generated by their brains, leading to inconsistencies in the labels [74, 168]. Besides, owing to the distinctive differences in each individual's brain, substantial variations exist in brain signals among different subjects [21, 176]. This diversity may result in evaluative discrepancies in labels annotated by domain experts, as different experts might assign different labels to the same EEG segment [88]. Such variations introduce an inconsistency in the labeled samples. Therefore, mitigating the significant influence of inconsistency issues in the training process and improving generalization ability became critical problems faced by the EEG analysis [123].

Self-supervised learning (SSL) has shown its superior performance in solving the challenges mentioned above, which leverages the intrinsic structure and information within data to train models without labels. Self-supervised learning designs a series of pretext tasks that differ from the final modeling target, generating *pseudo-labels* directly from unlabeled samples to train the model [40]. In the **Computer Vision (CV)** and **Natural Language Process (NLP)**, self-supervised learning has achieved tremendous success. In CV, SSL structure helps the model to learn the effective image representation through the pretext tasks such as image rotation [46], jigsaw [98], and reconstruction [55], which significantly improved the downstream task (specific target task that benefits from the pre-training, e.g., image classification and object detection) performance, sample efficiency, and mitigated the overfitting problem [66]. In NLP, the **mask-reconstruction (MAE)** and the prompt-answering pretext tasks help the language model to comprehensively understand textual context, enabling a series of functions of machine translation and conversation system [33]. Therefore, the strong representation ability and reliance on only a few number of labeled samples in the self-supervised learning paradigm highlight its potential as an effective training method, providing new insights and tools for addressing various complex problems in different domains.

Implementing the SSL frameworks in the EEG field is gaining more and more attraction among various researchers [112]. Figure 1 illustrates a typical SSL-integrated EEG analysis method. There have been certain studies investigating the combination of SSL with EEG analysis, which conduct the preliminary exploration of SSL to deal with temporal physiological signal-based tasks. Accordingly, this article comprehensively reviews the utilization of self-supervised learning for EEG analysis, which provides an in-depth exploration of the taxonomy, the pros and cons, and the development potential of the EEG-based SSL frameworks. In particular, we highlight recent efforts toward building large-scale pre-trained EEG foundation models and discuss their emerging role in enabling more generalizable and robust EEG analysis across tasks and datasets. We list the main contributions of this article as follows:

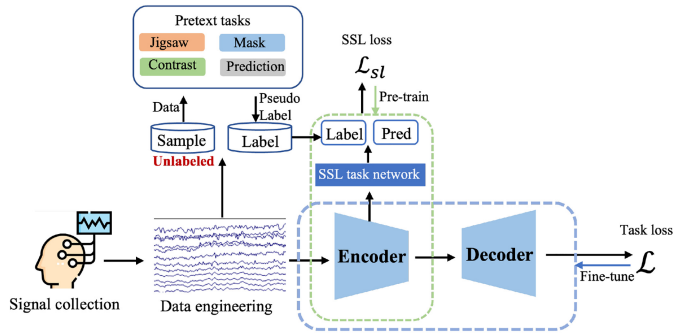


Fig. 1. The general process of SSL-integrated EEG analysis. The black arrows represent forward propagation, the green and blue arrows denote backpropagation based on pretext task loss and downstream task loss, respectively.

(1) Comprehensive review. This article provides a comprehensive up-to-date review of the self-supervised learning integrated EEG analysis methods. We analyze the technical details of different SSL approaches for EEG signals, including the type of pretext tasks, the mathematical description, the training mode, the downstream tasks, the SSL-driven EEG foundation models, and some simple summaries. By comparing different methods, we outline the general process and characteristics of the EEG-based SSL methods.

(2) Systematic and reasonable taxonomy. Following the classical taxonomy of the traditional self-supervised learning methods [50, 92], we rigorously categorize existing studies on self-supervised learning in EEG into four major classes: the prediction-based method, the generation-based method, the contrastive-based method, and the hybrid method.

(3) Future potential directions. We also analyze the pros and cons of various methods, identify the limitations of current works, and take into account the inherent characteristics of EEG data to indicate the potential directions for developing SSL-based EEG analysis methods.

2 Preliminary

This section provides a concise overview of traditional supervised EEG-analysis methods. In addition, we outline the form definition and mathematical description of the self-supervised learning frameworks proposed in other fields (e.g., CV, NLP), which serves as a preliminary for the EEG-based self-supervised method.

2.1 Supervised EEG Analysis

EEG signals have been widely studied to decode brain activity for addressing various real-world tasks. For instance, EEG have been used to recognize specific emotion [172], detect seizure [6], classify sleep stage [13], recognize motor imagery [7], decode visual or auditory information [71], and so on. Despite the manual analysis of EEG signals, machine learning and deep learning supervised methods have been widely adopted to analyze EEG signals, extract features, and complete specific tasks. Existing studies can be classified into two categories [142]: feature-driven and model-driven methods, where the feature-driven methods combine the handcrafted features with traditional machine learning classifiers to interpret EEG signals, and the model-driven methods construct end-to-end deep learning models to automatically extract task-related EEG features.

Feature-driven methods. The feature-driven methods use specific features extracted from EEG signals to guide the analysis process. In general, the feature-driven methods select handcrafted features that have been proven effective for the task according to the previous research by

neuroscientists [124]. By leveraging the selected features through traditional machine learning classifiers, the models can extract patterns, relationships, and insights in understanding EEG signals and brain activity. Various handcrafted features fed into different models are extensively applied to multiple tasks. For example, time domain features like the Hjorth parameter [99], the high order crossing [106], the statistical analysis features [135], and so on.; the frequency domain features like different independent frequency bands [29] generated through Fast Fourier Transform (a mathematical operation that transforms temporal signals into frequency domain) and differential entropy [38], and so on; the temporal frequency domain features which combined the frequency features with the time window to introduce the variation of frequency features over time [135]. Utilizing these manually engineered features as input, machine learning models have demonstrated a dependable performance in tasks such as emotion recognition, sleep stage classification, and motor imagery classification [63].

Model-driven methods. Different from the feature-driven methods, the model-driven methods rely on deep end-to-end models to interpret and analyze either raw temporal EEG data or high-dimensional EEG features [130]. These deep models are capable of capturing specific spatial-temporal patterns to infer underlying brain dynamics [42, 140], quantify brain activity [15, 120], and perform complex EEG-based classification or regression tasks [1, 30, 35, 107]. The existing model-driven methods predominantly represent supervised deep learning approaches that rely extensively on a substantial volume of training samples [83], which can be expressed as follows:

$$\theta^{d*} = \arg \min_{\theta^d} \mathcal{L}_{sl}(f_{\theta^d}(X), Y), \quad (1)$$

where θ^d represents the trainable parameters, and f_{θ^d} represents deep end-to-end models to extract features from EEG signals. $\mathcal{L}_{sl}$ denotes the supervised loss function, which measures the discrepancy between the predicted outcomes $f_{\theta^d}(X)$ and the ground truth labels Y . Owing to the powerful learning capabilities of deep learning and the availability of extensively labeled datasets, the performance of these models has been significantly enhanced across a wide range of complex EEG-related tasks [97, 128, 164].

The workflow of both feature-driven and model-driven methods follows a consistent structure, comprising modules such as data acquisition, feature extraction, and task implementation. This structure constitutes the standard pipeline of the EEG supervised learning process.

2.2 Overview of Self-supervised Learning

Self-supervised learning can extract effective representation from unlabeled samples instead of directly training end-to-end models through labeled samples, which has shown its superior performance in learning spatial images and sequential context representation in the fields of CV and NLP. In this part, we explain the terms of essential concepts, outline the mathematical definition of the SSL, and briefly divide the existing SSL frameworks into four distinct categories based on different pretext tasks.

Term explanation. We provide important definitions of terms to help further understand self-supervised learning.

- **Pretext task.** The pretext tasks $T = \{t_1, t_2, \dots, t_n\}$ refers to the learning objective or task designed to leverage the content or structure within unlabeled data to help the model learn knowledge and effective representations. The learned representations can then be transferred to downstream tasks with limited labeled data.
- **Pseudo-label.** The pseudo-label Y_p is the artificial label created based on the pretext tasks to train the model. The pseudo-labels serve as a form of supervision that guides the self-supervised learning process to extract specific features from unlabeled samples.

- **Downstream task.** The definition of downstream task d_t is the target or final task to be performed using the features or representations learned from the previous phase of training (by pretext tasks). The downstream task typically requires labeled samples to fine-tune the previous model to transfer the representation model to become more specific and task-focused.
- **Human-label.** The human-label Y refers to the labels for the downstream samples annotated by human experts.

Mathematical definition. The objective of self-supervised learning is to learn a function $\mathcal{F}(x) \rightarrow \mathbb{R}^d$ that maps the input instances X to a d -dimensional representation space $\mathbb{R}^d$, capturing essential features from unlabeled samples [171]. The frameworks of self-supervised learning are generally regarded as the **encoder-decoder** structure encompassing an encoder f_θ to generate representation and several decoders g to decode the representation to complete different tasks [24, 69]: **pretext task decoder** g_δ^p cascade with the encoder to accomplish pretext task and pre-train the model without external labels, and **downstream task decoder** g_ξ^d can recognize specific patterns in the representation to fine-tune the model adopted to downstream tasks. In general, the training paradigms of SSL can be summarized into three categories [116]: (1) pre-train mode; (2) joint-train mode; (3) linear evaluation mode.

The first mode is the pre-train mode, which employs pretext tasks to pre-train the representation model, followed by downstream tasks that fine-tune both the encoder and task-specific decoder to adapt the model for specific applications [23, 50]. The process can be expressed as follows:

$$\begin{aligned}\theta^*, \delta^* &= \arg \min_{\theta, \delta} \mathcal{L}_{pt} \left(g_\delta^p \left(f_\theta^p(X) \right), Y_p \right), \\ \theta^{**}, \xi^{**} &= \arg \min_{\theta^*, \xi^*} \mathcal{L}_{ft} \left(g_{\xi^*}^d \left(f_{\theta^*}^p(X) \right), Y \right),\end{aligned}\tag{2}$$

where $\mathcal{L}_{pt}$ and $\mathcal{L}_{ft}$ represent the loss of pretext task and downstream task, respectively. The encoder is trained by the pretext task and fine-tuned by the downstream task to first generate effective representations and then transfer the learned knowledge to specific tasks. The downstream task decoder is trained by the downstream task loss to fully leverage the representation for target task completion.

The second mode is the joint-train mode, where a combined loss function is constructed to leverage pretext and downstream tasks to jointly train the model [27]. Compared to other training modes, joint-train mode resembles a multi-task supervised learning approach, where the pretext tasks can still contribute effectively under fully supervised scenarios by collaboratively enhancing the training process [73]. The pretext task collaboratively explores the relevant knowledge for the downstream task and also serves as the regularization term to constrain the gradient during training, thereby mitigating the overfitting problem [132]. This mode can be expressed as follows:

$$\theta^*, \xi^* = \arg \min_{\theta, \delta, \xi} \alpha \mathcal{L} \left(g_\delta^p \left(f_\theta^p(X) \right), Y_p \right) + \beta \mathcal{L} \left(g_\xi^d \left(f_\theta^p(X) \right), Y \right),\tag{3}$$

where α and β are the hyper-parameter to balance different losses.

The third mode is the linear evaluation mode, which is similar to the pre-train mode, but the parameters of the encoder are frozen during the fine-tuning stage. This mode only fine-tunes the downstream task decoder to verify the generated representations' effectiveness [150]. Linear evaluation has become a widely adopted protocol for assessing the quality of representations learned

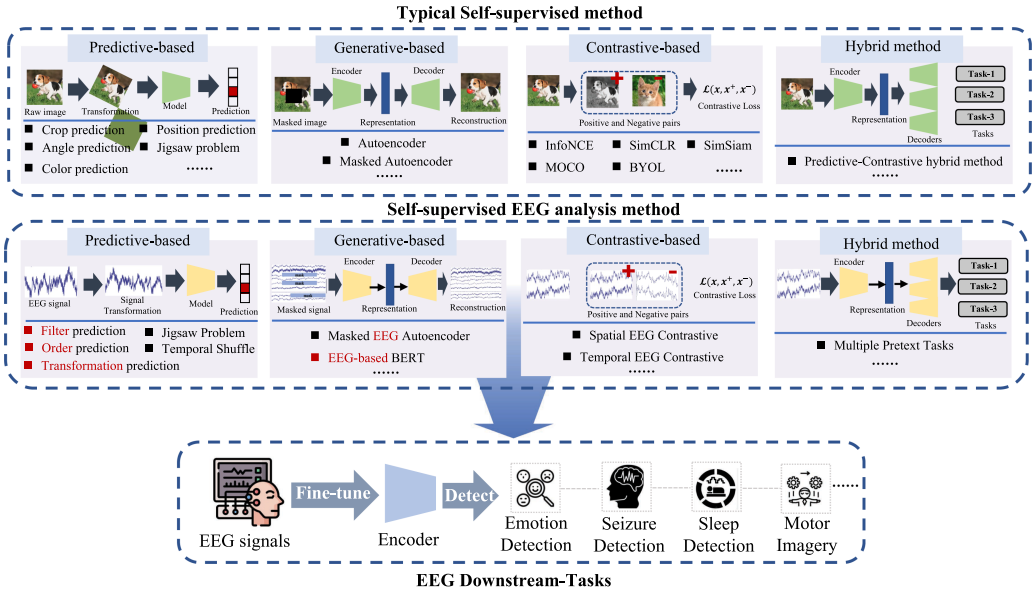


Fig. 2. The taxonomy of the typical self-supervised learning methods and self-supervised EEG analysis methods.

by self-supervised learning models [56, 81]. The process can be formulated as follows:

$$\begin{aligned} \theta^*, \delta^* &= \arg \min_{\theta, \delta} \mathcal{L}_{pt} \left(g_{\delta}^p \left(f_{\theta}^p(X) \right), Y_p \right), \\ \xi^* &= \arg \min_{\xi} \mathcal{L}_{ft} \left(g_{\xi}^d \left(f_{\theta^*}^p(X) \right), Y \right). \end{aligned} \quad (4)$$

Within the pretext task based taxonomy of SSL, we can categorize the SSL method into four types: predictive-based, generative-based, contrastive-based, and hybrid SSL method. Figure 2 demonstrates the general taxonomy of SSL, and the detailed explanations are as follows:

Predictive-based SSL method. The predictive-based SSL method creates classification pretext tasks that predict discrete pseudo labels derived from unlabeled data, enabling the learning of effective features. For instance, pretext tasks like predicting image rotation angles [46] and pixel colors [163] can force the model to extract spatial features and object boundaries in the images beneficial for downstream tasks such as object detection; pretext tasks like predicting the next sentence [33] can help the language model understand contextual correlation. Due to their simpler execution nature, predictive-based SSL methods are mostly easy to combine with traditional deep models, and the proficient performance in prediction tasks signifies that the model has mastered specific knowledge for downstream tasks [113, 118, 145].

Generative-based SSL method. The generative-based SSL method designs generation or reconstruction pretext tasks to capture contextual features and correlations to extract effective representations. The most widely used generative-based pretext task is the reconstruction task. This task begins by encoding the input sample into a distinctive representation, followed by a decoding process to reconstruct the original input. By making the input and output as similar as possible, the encoder can learn significant features to reconstruct the input, which are highly effective for target downstream tasks. For example, typical methods like autoencoder [159] have been investigated to extract representations from image and textual data. Recently, the mask-reconstruction

pretext task has replaced traditional reconstruction tasks to extract the contextual information from unlabeled samples [18, 55, 160]. This task masks part of the input samples and reconstructs the masked data through the contextual data, where the encoder is responsible for extracting features and representations, and the decoder is responsible for reconstructing the masked data. In vision tasks, the masked autoencoder [55] can extract spatial contextual features from unlabeled samples for downstream classification and segmentation. In language tasks, the BERT [33] model captures token-level context correlation information, which greatly improves the performance of subsequent tasks such as machine translation and text generation.

Contrastive-based SSL method. The contrastive-based SSL method adopts the “comparison” technique, which encourages similar data points to be closer in the representation space while pushing dissimilar data points apart. Augmentation methods are important in the contrastive-based SSL: input samples are augmented to create negative and positive sample pairs, where the positive pairs represent the similar samples, and negative pairs refer to the vastly dissimilar samples [139]. By optimizing the designed contrastive loss, the model minimizes the distance between positive pairs and maximizes the distance between negative pairs to extract identical features and transferable representations. Based on the theory of information bottleneck [134] and mutual information [78], InfoNCE loss [60] is proposed to efficiently learn representations where positive pairs are closer together in the feature space compared to negative pairs. Besides, SimCLR [23], MoCo [56], and other contrastive learning methods have become important frameworks driving the development of computer vision.

Hybrid SSL method. The hybrid SSL method combines multiple SSL techniques or tasks to create a powerful framework for learning representations [125]. The main idea is to leverage the strengths of different pretext tasks to capture diverse and informative features from unlabeled samples [34, 118]. The weighted fusion of losses from multiple pretext tasks enables the model to grasp multi-dimensional knowledge [17, 89]. It is particularly valuable when data is heterogeneous or a single pretext task may not capture all the relevant information in the unlabeled samples [92].

Following the taxonomy of typical SSL frameworks in vision and language fields, this article categorizes self-supervised EEG analysis methods into predictive, generative, contrastive, and hybrid frameworks. Comprehensive summaries for different methods are provided from Section 3 to Section 6. The structure of this survey can be visualized in Figure 3.

3 Predictive-based SSL EEG Analysis Method

The predictive-based SSL EEG analysis method aims at designing and executing classification to acquire domain-specific knowledge beneficial for various downstream tasks. Multi-channel EEG signals present distinctive characteristics, including high temporal density, pronounced temporal dependencies, and intricate inter-channel correlations, indicating the presence of critical features within the temporal, frequency, and spatial domains of EEG data. Sequentially, pretext tasks are implemented to distinguish EEG samples that are augmented through temporal, frequency, and spatial processing to acquire features from different domains. Therefore, we can categorize the existing studies into three subcategories: (1) *spatial predictive methods*, (2) *temporal predictive methods*, and (3) *transformation predictive methods*. The typical frameworks of three kinds of methods are demonstrated in Figure 4 and Figure 5, and the summary of existing works is listed in Table 1.

3.1 Spatial Predictive Method

The spatial predictive method draws inspiration from SSL in the image domain, establishing local or global spatial-structure-related pretext tasks to help the model comprehend spatial contextual information. Figure 4(a) shows the typical spatial predictive framework for EEG analysis, and

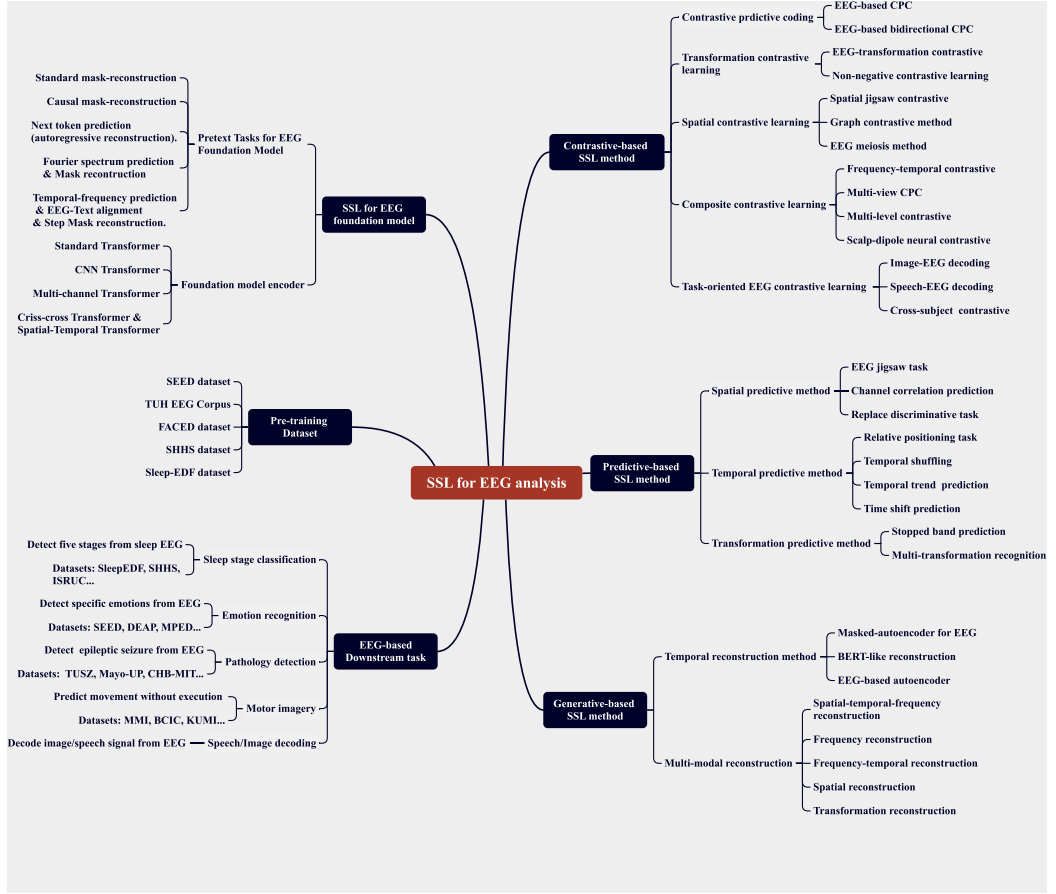


Fig. 3. Categories of Self-supervised learning for EEG analysis.

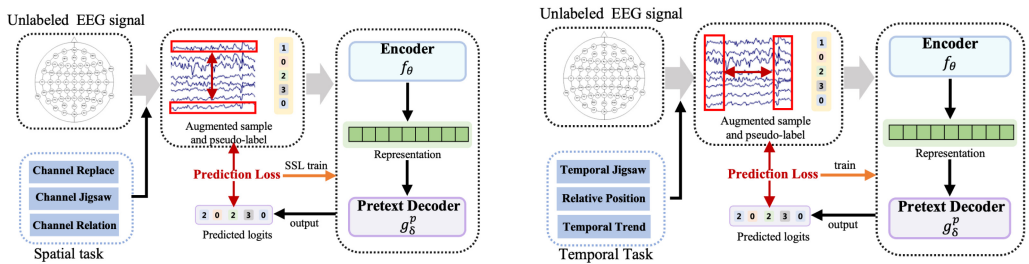


Fig. 4. The comparison of spatial predictive and temporal predictive SSL EEG analysis methods.

different methods have been investigated to extract channel correlation and brain structure, which are listed as follows:

EEG jigsaw task [86] is analogous to the image jigsaw pretext task in CV. The EEG jigsaw task involves the random shuffling of EEG channels, followed by the expectation that the model

can reconstruct the original sequence of the scrambled channels or predict the order in which they were shuffled. For example, assume the raw EEG data $X_{sp} \in \mathbb{R}^{c \times t}$, where c is the number of channels and t is the number of sampling points. The random shuffling operation then produces a permuted EEG matrix $X_{sp}^* \in \mathbb{R}$, where the temporal information remains unchanged, but the channel order has been shuffled. The loss function of the jigsaw task can be described as follows:

$$\mathcal{L}(X_{sp}^*, Y^*) = - \sum_{i=1}^N Y_i^* \log \left(g_{\delta}^p \left(f_{\theta} \left(X_{sp_i}^* \right) \right) \right), \quad (5)$$

where Y^* represents the one-hot pseudo-label (channel order), and N represents the batch size. The loss function calculates the cross-entropy between the predicted order of the shuffled EEG sample and its corresponding label. By minimizing this pretext loss, researchers believed the model can capture spatial features related to the distribution of multi-channel EEG signals across the cortical regions of the brain [86], which are closely related to downstream tasks such as emotion recognition and seizure detection.

Channel correlation prediction [17] is designed to realize the spatial correlation between different channels. Researchers proposed that the time delay exists in the propagation of EEG signals between channels in distinct brain regions [17]. EEG signals will experience delays when propagating from one region to a distant one, and exploring these features enables the model to understand the activation modes and information exchange patterns of brain activity. In this task, pseudo labels are generated from signal correlation between channels, which can be calculated as follows:

$$Y(i, j, t_1, t_2) = \begin{cases} 1, & \text{cossim}(X_i(t_1), X_j(t_2)) \geq \kappa \\ 0, & \text{cossim}(X_i(t_1), X_j(t_2)) < \kappa \end{cases} \quad (6)$$

This function calculates the cosine similarity between the i th channel and the j th channel at time slices t_1 and t_2 . The *cossim* represents the cosine similarity, and the κ is the predefined threshold to determine whether the two slices are correlated and assign the pseudo labels. The binary cross-entropy loss can be used as prediction loss to pre-train the model, where the predictions are generated by the encoder-pretext task decoder structure: $g_{\delta}^p(f_{\theta}(X_1, X_2))$.

Replace discriminative task [17] is the binary classification task to extract channel-specific differential features by identifying distinct components from different channels. In this task, a random replacement is performed to replace a certain percentage $p_r\%$ of original EEG signal X_i with signal $\bar{X}_i$ sampled at any channels and time slices. The pseudo labels are constructed to indicate whether the current samples have been replaced, which can be described as follows:

$$Y(X_i) = \begin{cases} 1, & f_I(X_i, i) = 0 \\ 0, & f_I(X_i, i) \neq 0 \end{cases} \quad (7)$$

where $f_I(X_i, i)$ is the function to judge whether the signal X_i has been replaced or not. Subsequently, by minimizing the binary cross-entropy pretext task loss, the model learns the distinctive spatial features of different channels and retains essential information beneficial for various downstream tasks.

3.2 Temporal Predictive Method

The temporal predictive methods aim at capturing the temporal correlation and sequential dependencies in EEG signals. As the temporal physiological signal, temporal characteristics play an important role in various EEG-based tasks. Figure 4(b) shows a typical framework of temporal predictive SSL for EEG analysis, and different temporal predictive pretext tasks have been proposed to investigate potential temporal information, which are summarized as follows:

Relative positioning task is the temporal predictive method to distinguish whether two different EEG segments are close or distinct in time dimension [12]. This task firstly constructs an EEG pair $X_{t_i}, X_{t'_i} \in \mathbb{R}^{c \times t}$ represent two sampled EEG segments. EEG representations are expected to evolve slowly over time, meaning that temporally adjacent segments convey similar information, while those farther apart tend to exhibit greater dissimilarity [12]. The duration parameter τ_{pos} controls the duration of positive context. For two EEG segments X_{t_i} and $X_{t'_i}$, the temporal interval $|t_i - t'_i| \leq \tau_{pos}$ indicates that these segments are within positive duration, sharing common underlying labels. Therefore, the pseudo labels of the relative positioning task can be constructed as follows:

$$Y(X_{t_i}, X_{t'_i}) = \begin{cases} 1, & |t_i - t'_i| \leq \tau_{pos} \\ -1, & |t_i - t'_i| > \tau_{pos} \end{cases}, \quad (8)$$

where -1 and 1 represent samples of negative and positive duration. The training sample $S_N = \{(X_{t_i}, X_{t'_i}), Y((X_{t_i}, X_{t'_i}))\}$ can be used to train the model with binary classification loss to capture temporal information. This method has been widely used for continuous EEG classification tasks such as sleep stage classification [12, 48, 147].

Temporal Shuffling is considered as the variation of the relative positioning task [12]. The temporal shuffling task first samples two different EEG segments $X_{t_i}, X_{t'_i}$ from positive duration, and then samples another EEG segment $X_{t''_i}$ between the first two segments or in the negative duration. Three different segments form the triplet $(X_{t_i}, X_{t'_i}, X_{t''_i})$. The shuffling operation is performed to permute the order of the segments in the triplet randomly. The pseudo labels indicating whether the triplet has been shuffled can be constructed: 0 for the shuffled triplet and 1 for the normal triplet. Then, the model learns to distinguish whether the triplet has been shuffled through the calculated differential features (absolute differences) between segments, which can be calculated as follows:

$$D(X_{t_i}, X_{t'_i}, X_{t''_i}) = \text{concat} \left(\left| f_\theta(X_{t_i}) - f_\theta(X_{t'_i}) \right|, \left| f_\theta(X_{t'_i}) - f_\theta(X_{t''_i}) \right| \right), \quad (9)$$

where *concat* is the vector concatenation operation. The model conducts shuffling classification by utilizing differential encoded information from different segments as features, which can help to comprehend temporal dependencies within EEG signals. Besides, another temporal shuffling method proposed by [102] divides the EEG slice into three equidistant sequences, then randomly shuffles the order of sequences to form the shuffled sample. The model is asked to predict the order of input shuffled samples to capture temporal correlation. Therefore, for shuffled EEG signals, both binary classification (predict whether they have been shuffled) and multi-class classification (predict the order of shuffled signals) can serve as the pretext task to extract temporal features of physiological signals at different granularities.

Temporal trend prediction [76] is a task to identify the potential trends of EEG to capture short-term and long-term dynamic patterns. This task divides the EEG signal into three categories according to its temporal characteristics: stationary, trendstationary, and cyclostationary. By learning how to identify temporal trends, the model can understand the temporal relationships within signals and capture both global and local essential waveform information to generate the temporally enriched representations, which can benefit various downstream tasks, like sleep stage classification.

Time shift prediction [2] is a task to predict the time shift performed to the EEG signals by contrasting the differences in features between the raw EEG signal and shifted signals. In this task, the raw EEG signal X_{t_i} and augmented EEG signal $X_{t_i+\rho}$ result from ρ -step time shifts

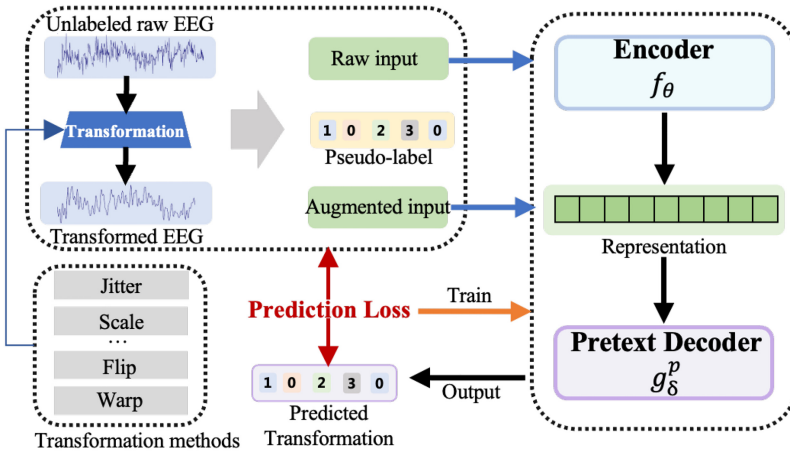


Fig. 5. The general process of transformation predictive method for EEG analysis. Different signal transformation techniques are applied to EEG signals to generate augmented samples and pseudo labels. The model can capture critical signal-level features for downstream tasks by correctly predicting the transformation method.

applied to the raw EEG signal. The raw signal and shifted signal are encoded into representations, and the pretext task uses a classification method to analyze the difference between the two representations and classify how much the raw signal was shifted. By minimizing the classification loss, the encoder can learn the temporally aligned features and dependencies within EEG signals, generating the representation containing rich time information significantly beneficial for long-term EEG tasks like clinical monitoring.

3.3 Transformation Predictive Method

Figure 5 shows the general process of the transformation predictive method. This task aims at predicting specific augmentations applied to the EEG signals to learn signal-related features in the time-frequency domain. Different EEG transformation techniques employed to augment EEG samples for pretext tasks can be listed as follows:

Stopped band prediction randomly removes specific frequency bands from EEG signals and forces the model to predict the index of the removed channel to learn frequency-related features [70]. EEG signals comprise information from multiple frequency bands, with essential information concentrated within the frequency range of 1 to 50 Hz, encompassing five independent frequency bands: δ (0.5–4 Hz), θ (4–8 Hz), α (8–12 Hz), β (12–30 Hz), and γ (30–50 Hz). This task transforms the EEG signal from time to the frequency domain and remaps the signal to the time domain after the random removal of a specific frequency band, and the pseudo labels $Y(x_i) \in [0, 1, 2, 3, 4]$ are set representing the index of the removed band. By forcing the model to predict the stopped band through encoded representation $f_\theta(X_i)$, the encoder f_θ can learn efficient frequency correlation and features to form the temporal-frequency representation.

Multi-transformation recognition aims at predicting the transformation technique used to augment EEG signals to extract fine-grained signal features and form effective representations [138]. In this task, EEG signals are augmented through one transformation technique, and the model is asked to recognize the transformation methods. The common encoder f_θ extracts features from augmented EEG signals and encodes them into representations, with multiple binary classifiers to recognize different transformation methods. Each classifier corresponds to

a specific transformation method to determine its occurrence. Six transformation methods are proposed to be recognized:

- (1) *Noise Adding*. Adding random noise NS_i to the raw signal sampled from the Gaussian distribution: $NS_i \sim N(\mu, \sigma^2)$. The noise NS_i is directly added to the original signal X_i , resulting in a noise-augmented signal X_i^{ns} .
- (2) *Scale transformation* alters the waveform of the EEG signals. The amplitude of EEG signal is scaled through a scale factor α , where the scale-augmented signal can be expressed as $X_i^{st} = \alpha * X_i$.
- (3) *Horizontal flipping*. This transformation method directly flips EEG signal horizontally. The horizontal-augmented signal can be expressed as $X_i^h = -X_i$.
- (4) *Vertical flipping* flips EEG signal (each sample) vertically. The vertical-augmented signal can be described as $X_i^v = flip(X_i)$, where the *flip* represents the segment's vertical symmetric flip.
- (5) *Temporal dislocation* is consistent with the temporal shuffling in predictive methods. This method divides EEG segments into sub-segments and randomly shuffles these sub-segments to form the dislocation-augmented signal X_i^{td} .
- (6) *Time warping* method randomly stretches and compresses sub-segments to form the augmented samples. This method randomly selects sub-segments to stretch and compress, with the recombination method to reassemble all the sub-segments and construct the warping-augmented signal X_i^{tw} with the same dimension as the original EEG signal.

By recognizing different transformation techniques, the model can generate the representation that captures temporal dependencies, frequency correlation, and time-frequency correspondences within EEG signals from unlabeled samples.

3.4 Section Discussion

This section extensively reviews the predictive-based EEG analysis methods. Specifically, we categorize the predictive methods into three subcategories: spatial predictive, temporal predictive, and transformation predictive methods. The spatial predictive tasks focus on exploring the channel-correlation features, emphasizing spatial relationships among EEG channels. In contrast, the temporal predictive tasks involve incorporating rich temporal dependency features, time-correlation features, and consistent temporal information into the representation. The transformation predictive task can help the model to extract temporal-frequency aligned features by recognizing typical signal transformation techniques. These pretext tasks are computationally efficient to implement, with the pretext task decoder g_δ^p can be simple forward neural networks or the traditional machine learning classifiers [12, 173]. The predictive-based SSL EEG analysis methods are summarized in Table 1, which highlights the diversity of pretext tasks, backbone architectures, downstream tasks, and training modes. Notably, transformation predictive methods dominate the field, demonstrating their effectiveness across various tasks like seizure detection, emotion recognition, and pathology classification. CNN is the most commonly used backbone, reflecting its strong adaptability to EEG data. Furthermore, training modes such as **pre-training (PT)** and **fine-tuning (FT)** are widely utilized to validate the SSL framework, emphasizing the importance of full-parameter transfer in EEG-based SSL approaches. Some methods, like GMSS, leverage **joint training (JT)**, indicating the potential benefits of multi-task learning frameworks. In summary, predictive-based SSL methods provide a robust foundation for extracting meaningful spatial-temporal representations from EEG data, enabling advancements in both foundational research and practical applications.

Table 1. The Summarization of Predictive-based EEG Analysis Self-supervised Learning Method

Approach	Type of Pretext Method	Detailed Method	Backbone	Downstream Tasks	Training Mode
EEG scaling SSL [151]	Transformation predictive	Scaling prediction	SVM	Epileptic classification	PT
GMSS [86]	Spatial predictive	EEG jigsaw	GNN	Emotion recognition	PT&JT&LE
Transformation SSL [138]	Transformation predictive	Multi-transformation	CNN	Emotion recognition	LE
Task-agnostic SSL [104]	Transformation predictive	Multi-transformation	CNN	Seizure and Motor imagery	LE
Temporal EEG SSL [48]	Temporal predictive	Relative position	-	Sleep and Pathology prediction	PT
SSL-EED AD [173]	Transformation predictive	Noise classification	CNN,SVM	Pathology prediction	LE
Speech-EEG SSL [2]	Temporal predictive	Temporal-shift	CNN	Speech decoding	LE
SSL MI-EEG [102]	Temporal predictive	Temporal shuffling	CNN	Motor imagery	PT
MtCLSS [87]	Transformation predictive	Multi-transformation	CNN	Pediatric sleep classification	JT
MM Emotion [96]	Transformation predictive	Multi-transformation	CNN	Emotion recognition	PT
SSTSC [147]	Transformation predictive	Relative position	CNN	Seizure detection	PT
Clinical EEG SSL [12]	Temporal predictive	Relative position Temporal shuffling	CNN	Sleep classification Pathology classification	PT
SSL for sleep EEG [11]	Temporal predictive	Relative position Temporal shuffling	CNN	Sleep classification	PT
EEG-oriented SSL [76]	Transformation predictive Temporal predictive	Band-stop prediction Temporal-trend	CNN	Sleep and Pathology Motor imagery	PT
Robust EEG SSL [70]	Transformation predictive Temporal predictive	Band-stop prediction Temporal-trend	CNN	Sleep and Pathology	PT
MBrain [17]	Spatial predictive	Channel correlation Replace discriminative	CNN,LSTM	Seizure detection	PT

In the “training mode” column, “PT” represents pre-training and fine-tune mode, “LE” represents linear evaluation mode, and “JT” represents joint-training mode.

4 Generative-based SSL EEG Analysis Method

Different from predictive methods, generative-based SSL EEG analysis methods are more complex and challenging. The critical terms of this method are “*Reconstruction*” and “*Generation*,” where the fine-grained correlation and features can be captured through this pretext task. “*Reconstruction*” means reconstructing the masked or transformed samples to learn effective representation, and “*Generation*” means generating specific context to train the model to learn specific semantics. For instance, by reconstructing the masked image, the model can capture spatial correlation and semantics [55]; by generating the contextual information of language tokens, the model can understand the language semantics for different downstream tasks [119]. In the EEG analysis, the generative-based SSL method adopts signal reconstruction as the pretext tasks, which can be categorized into two independent sub-categories according to the task target: (1) *Temporal reconstruction task*, (2) *Multi-domain reconstruction task*. The typical frameworks are demonstrated in Figure 6, and the summary of existing works is listed in Table 2.

4.1 Temporal Reconstruction Task

The framework of the temporal reconstruction task is shown in Figure 6(a). It is inspired by the autoencoder method [159], which reconstructs the input data to capture contextual features without requiring human-labeled samples. The reconstruction task enables the encoder to learn fine-grained input correlation, which can generate representations containing rich contextual information. EEG signals are serialized temporal physiological data. This characteristic makes them well-suited for the temporal reconstruction pretext task, which aims at learning contextual correlations within the signal, enhance the understanding of temporal dependencies, and provide effective representations for various EEG-based downstream tasks. Different temporal reconstruction tasks are listed as follows:

EEG-based autoencoder [64] is the adaptation of autoencoder for EEG analysis [59]. In this method, EEG signals are encoded into low-dimensional representation by the encoder f_θ , and the

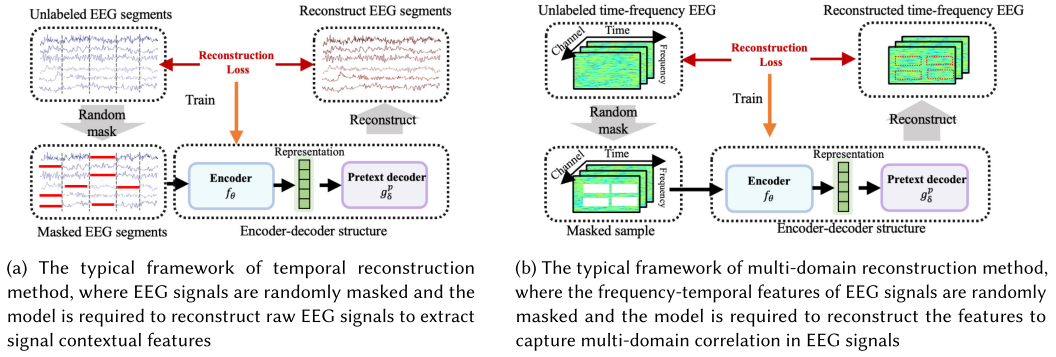


Fig. 6. The frameworks of temporal reconstruction and multi-domain reconstruction SSL EEG analysis methods.

low-dimensional representation is then used to reconstruct the original signal through the pretext task decoder g_θ^p symmetrical to the encoder. The encoder is responsible for preserving critical EEG signal information, while the decoder is responsible for reconstructing the EEG signal from the generated representation. The reconstruction loss can be calculated as follows:

$$\mathcal{L}(X, X^*) = \frac{1}{\text{len}(X)} \|X - X^*\|_1, \quad (10)$$

where $\text{len}(X)$ represents the length of input signal and $\|\cdot\|_1$ represents the L1-norm. X is the EEG signal and X^* is the reconstructed signal. By minimizing the difference between the original and reconstructed signals, the encoder preserves critical information necessary for signal recovery, which can be considered signal compression.

Signal-level mask-reconstruction (signal MAE) is the typical mask-reconstruction method to capture temporal signal correlation for reconstructing the masked segments [28]. In this framework, multi-channel EEG signals are first encoded into temporal embeddings through the 1D convolution block similar to the famous wave2vec [8, 121] algorithm. The high-dimensional EEG signals X are downsampled and compressed into low-dimensional feature embeddings $Z = \{z_1, z_2, z_3, \dots, z_k\}$ arranged in temporal order, and the stride of the convolution block determines the number of input time steps to the encoder. The mask M_i is generated to randomly replace parts of the information in embedding Z , creating the masked embedding Z^m with local information dropout. The transformer encoders are then applied to extract bidirectional temporal correlation between input slices and output the signal representation $Re = \{re_1, re_2, \dots, re_k\}$. The convolution block and Transformer encoder are cascaded as the encoder f_θ to generate signal representation, and followed by the linear and convolution layer as the pretext task decoder g_θ^p to reconstruct the raw signal X through masked signal representation Re , the training process can be described by minimizing cosine similarity loss shown as follows:

$$\mathcal{L}(X_m, X) = 1 - \frac{X_m \cdot X}{|X_m||X|}, \quad (11)$$

where X_m is the reconstructed EEG signal. This method reconstructs raw signal through the masked signal representation, where the original EEG signals serve as the pseudo-labels. In this architecture, encoder f_θ is responsible for mining temporal correlation and preserving critical signal information while the pretext decoder g_θ^p is responsible for reconstructing original EEG signals. Therefore, this framework forces the encoder to generate representations containing fine-grained

correlation and signal critical information, which exhibits strong expressiveness, generalization, and applicability across various EEG-based tasks [156, 174].

Embedding-level mask-reconstruction (embedding MAE) is another mask-reconstruction method that is inspired by the BERT model[33] in the language domain to fuse the contextual relationship into the representation [77]. Similar to the MAE framework mentioned above, EEG signals are first transformed into temporal embeddings through the convolution block and then encoded by the transformer encoder to generate signal representations, followed by the pretext task decoder to accomplish the reconstruction task. However, different from the signal MAE, this task is based on the embedding-level reconstruction: the transformed EEG embedding Z is randomly masked by the generated mask vector, where z^* represents the randomly selected embeddings to be masked. The pretext task decoder is required to **reconstruct the masked embedding rather than EEG raw signals**. The contrastive loss function is designed to make the reconstructed embedding z_{pre}^* to be as similar as possible to the original unmasked embedding z^* while keeping it as dissimilar as possible to the remaining embeddings, which can be calculated as follows:

$$\mathcal{L}(z^*, z_{pre}^*) = -\log \frac{\exp(\text{sim}(z_{pre}^*, z^*)/\eta)}{\sum_{z_{r_i} \in Z} \exp(\text{sim}(z_{pre}^*, z_{r_i})/\eta)}, \quad (12)$$

where z_{r_i} is the negative sample obtained by random sampling from the contextual embeddings, sim represents cosine similarity to measure the distance between the reconstructed and original embeddings, and η is the temperature parameter to control the contrastive loss. Compared with the MAE to reconstruct the original signals, the embedding-level reconstruction is simpler, with fewer parameters to capture critical contextual information precisely and understand EEG embedding temporal relationships. However, this process may result in the loss of some original signal information. The combination of the transformation encoder can generate representations for various downstream tasks.

4.2 Multi-domain Reconstruction Method

The multi-domain reconstruction method extends the EEG-based MAE to multiple domains (signal, spatial, and frequency), which can be shown in Figure 6(b). Different from the temporal reconstruction methods, this method achieves collaborative and mutual reconstruction across different domains to extract spatial-temporal-frequency aligned and complementary features in the EEG signal, generating more powerful and general representations adapted to different tasks. Detailed explanations of multi-domain reconstruction methods are as follows:

Spatial-temporal-frequency reconstruction (STF MAE) conducts the synergistic reconstruction task in the temporal-frequency-spatial domain to extract integrated EEG features [20]. The idea of the synergistic reconstruction task is inspired by the time-frequency analysis method [115]: the temporal analysis method [10, 126] investigates the patterns of EEG amplitude changes over time, while the frequency analysis method [54] studies the frequency energy distribution within EEG signals. The time-frequency analysis method utilizes the sliding time window to investigate temporal changes in frequency spectral features [162]. Based on the time-frequency analysis method, this task constructs a 3D matrix as the feature of the EEG signal. Through the **continuous wavelet transform (CWT)** [3], EEG signals are transformed into a 3D frequency-spatial-temporal matrix $X \in \mathbb{R}^{c \times t_n \times f_r}$, where c is the channel number, t_n is the number of temporal slices, and f_r represents frequency feature resolution. This 3D matrix can be considered time-frequency features (2D image) with multiple channels. Inspired by the image MAE, the EEG feature matrix is divided into different patches and randomly masked with the mask patch m_p to generate the masked matrix X^* , the encoder-decoder structure utilizing **vision-transformer (ViT)** [36] as the backbone is designed to reconstruct the EEG feature matrix. The **mean squared error (MSE)** can be used

to train the model, which is defined as follows:

$$\mathcal{L}(X^m, X^{pre}) = E(X^{pre} - X^m)^2 = \frac{1}{n_m} \sum_{i=1}^{n_m} (X_i^{pre} - X_i^m)^2, \quad (13)$$

where n_m is the dimension of the masked features X^m , and X^{pre} is the reconstructed features generated by the encoder and pretext task decoder: $X^{pre} = g_\delta^p(f_\theta(X^*))$. By minimizing the MSE loss, encoder f_θ fuses spatial-temporal-frequency contextual correlation into representation, and decoder g_δ^p is learned to reconstruct the original EEG feature matrix based on the representation. The generated representations contain multi-domain correlation and features, which exhibit greater expressive ability for **Autism Spectrum Disorder (ASD)** detection and emotion recognition tasks [84].

Frequency mask-reconstruction (frequency MAE) conducts mask-reconstruction task in different frequency bands to capture frequency features, long-term dependencies, and critical time-frequency correlated information [105]. Initially, the EEG signal undergoes two distinct augmentations: (1). The EEG signal is directly embedded into the patch sequence through division, linear projection, and flattening operation, representing the EEG temporal patch sequence. (2). The EEG signal is transformed into six independent frequency bands (0–4 Hz, 4–8 Hz, 8–18 Hz, 16–32 Hz, 32–64 Hz, and other frequencies), representing the EEG frequency patch sequences. The definitions of these frequency bands differ from the classical definitions due to limitations in data sampling equipment and the customized experimental definitions adopted by the authors. 10% of the frequency patch sequences are randomly masked, followed by six independent ViT-based encoders to generate representations for all frequency bands (one encoder corresponds to one frequency band). Six independent ViT-based decoders are sequentially used to reconstruct the frequency patch sequences. Differently, the target of the pretext task is to minimize the difference between the summation of all reconstructed frequency sequences and the temporal patch sequence, which can be calculated similarly to Equation (15). This task reconstructs temporal information from masked frequency patch sequences. By doing so, it helps the model align temporal and frequency information within the EEG signals and understand their correlations. As a result, the model can generate high-dimensional representations with rich time-frequency coherent features, providing valuable EEG features for various EEG-based tasks.

Frequency-temporal reconstruction (FT MAE) is the framework to reconstruct the masked EEG representations in the frequency and time domain [144]. This framework transforms EEG signals into discrete patches through the non-overlapping 1D-CNN, with some patches randomly masked with ratio r . The ViT-based encoder is subsequently employed to generate representations, followed by a symmetric decoder to reconstruct the masked patches. Two reconstruction methods are proposed in the framework: the first is the **spatiotemporal domain reconstruction**, where the decoder reconstructs the masked patches directly, with the MSE loss function to train the model and capture the temporal correlations in EEG signal. The second is the **Fourier domain reconstruction**, which reconstructs masked patches in the frequency domain. Through the **Discrete Fourier Transform (DFT)** [9], EEG signals can be transformed from the time domain to the frequency domain, allowing the calculation of their amplitude and phase as frequency domain representations. Researchers believe that the study of both magnitude and phase is important [144]. In the EEG signal, the magnitude and phase are highly related to the phase synchronization information between neurons, which can help reveal the synchrony and information transmission between different brain regions. The Fourier domain reconstruction task predicts the magnitude and phase sequence of masked EEG patches, which are then reconstructed through the inverse Fourier transform. The mean squared error can measure the difference between the original patches and

those reconstructed by magnitude and phase. Encoder f_θ can understand the correlation between spectrum features and temporal signal and capture critical neuron activity knowledge through this task.

Spatial reconstruction (Spatial MAE) aims at learning the spatial correlation between different channels in EEG signal [61]. In this framework, the correlation between EEG channels can be defined using a graph structure $\mathcal{G} = (\mathcal{A}, \mathcal{X})$, where $\mathcal{X} \in \mathbb{R}^{c \times n}$ represents the node feature matrix in the graph (each channel corresponds to a node on the graph) and $\mathcal{A} \in \mathbb{R}^{c \times c}$ is the adjacency matrix representing the connectivity between nodes. The graph structure can be calculated through channel spatial distance and correlation. In the framework, the subgraph $\mathcal{G}^s$ is sampled containing n_s nodes and their connectivity graph structure. For the sampled sub-graph, the feature of a random node is masked and then reconstructed by the model through adjacent node features and graph structure, which can train the model to capture the spatial correlations. The **graph neural network (GNN)** [146] is used as the backbone for the encoder and decoder to deal with topological graph data, and the MSE is the loss to measure the node reconstruction performance. By reconstructing the graph node, the generated representation enables a deeper exploration of spatial features and channel correlation, which is valuable for tasks that require high spatial resolution in EEG (such as visual decoding).

Transformation reconstruction aims at reconstructing EEG signals after different signal augmentations to preserve critical signal-related information [31]. The model reconstructs the EEG signal after the following signal augmentations: (1) *Signal jitter*, where EEG samples are added with random noise: $x_s = x + s$. (2) *Random sample*, where some points in the temporal EEG signal are replaced by the average value of neighbor points. This transformation can be considered as the *smoothing* operation. (3) *Channel removal*, where a specific channel in the EEG signal is removed to be reconstructed. (4) *Window replace*, where EEG signals in randomly selected time windows are replaced by dummy value zero. (5) *Jitter in time windows*, where the signal in the randomly selected time window is corrupted by noise. The model can fuse temporal correlation, spatial correlation, and transformation features into representation for downstream tasks by reconstructing raw signals from various augmentations in the pre-training process.

4.3 Section Discussion

This section reviews the generative-based SSL EEG analysis methods, which leverage complex generative pretext tasks to train the encoder to capture effective signal features for downstream tasks. Existing approaches are classified into two subcategories: (1) The temporal reconstruction task that masks part of the temporal signal and requires the model to reconstruct. (2) The multi-domain reconstruction task that masks temporal-frequency features and requires the model to reconstruct. The generative-based SSL EEG analysis method is summarized in Table 2, highlighting significant advancements in the field. Both temporal reconstruction and multi-domain reconstruction tasks have been extensively studied and validated. The temporal reconstruction pretext tasks have demonstrated extensive applicability across various EEG-related tasks, including speech decoding, sleep stage classification, and emotion recognition. This indicates that the **Masked Autoencoder (MAE)** architecture, which excels at extracting contextual information in vision and language models [57], is equally effective in capturing complex EEG patterns. Multi-domain reconstruction pretext tasks, exemplified by methods such as EEGMAE and Wavelet2vec, further highlight the potential of leveraging diverse features—including spatial, frequency, and temporal characteristics—for addressing complex downstream tasks, such as ASD classification and seizure detection. Transformer and Vision Transformer architectures are frequently employed as backbone encoders, reflecting an emerging trend in using advanced models to extract complex features. Regarding training modes, pre-training and fine-tuning remain the most common,

Table 2. The Summarization of Generative-based EEG Analysis Self-supervised Learning Method

Approach	Sub-category	Detailed Method	Backbone	Downstream Tasks	Training Mode
BENDR[77]	Temporal reconstruction	Embedding MAE	CNN&Transformer	Multiple tasks	PT&LE
EEG-CGS[61]	Multi-domain reconstruction	Spatial MAE	GNN	Seizure analysis	PT
CEMOE[80]	Temporal reconstruction	Signal MAE	Transformer	Emotion recognition	PT
Eeg2vec[175]	Temporal reconstruction	Embedding MAE	CNN&Transformer	Speech decoding	LE
MAEEG[28]	Temporal reconstruction	Signal MAE	Transformer	Sleep classification	PT&LE
Cognitive MAE[109]	Temporal reconstruction	Embedding reconstruction	CNN&Transformer	Cognitive-load classification	PT&LE
MV-SSTMA[84]	Multi-domain reconstruction	STF MAE	CNN&Transformer	Emotion recognition	PT
EpilepsyNet[64]	Temporal reconstruction	EEG-based autoencoder	CNN	Epileptic classification	JT
EEGMAE[20]	Multi-domain reconstruction	STF MAE	ViT	ASD classification	LE&PT
brain2vec[82]	Temporal reconstruction	Embedding MAE	CNN&Transformer	Speech decoding	PT&LE
Wavelet2vec [105]	Multi-domain reconstruction	FT MAE	ViT	Seizure detection	PT
CRT [166]	Multi-domain reconstruction	STF MAE	Transformer	Sleep classification	LE
Neuro2vec [144]	Multi-domain reconstruction	FT MAE	Transformer	Seizure and sleep	PT&LE
SAE-EEG[90]	Temporal reconstruction	EEG-based autoencoder	CNN	Emotion recognition	PT
AE-CDNN[141]	Temporal reconstruction	EEG-based autoencoder	CNN	Seizure detection	LE
MI-AE[94]	Temporal reconstruction	EEG-based autoencoder	CNN	Motor imagery	LE

“PT” represents pre-training and fine-tune mode, “LE” represents linear evaluation mode, and “JT” represents joint-training mode.

though several methods adopt a combination of PT and linear evaluation to evaluate the quality of pre-trained EEG representations. We summarized three key trends in the generative-based SSL EEG analysis method: (1) The integration of spatial and temporal MAE methods, suggests a trend toward incorporating signal structure in pretext tasks to improve performance on downstream applications. (2) Complex sequence-to-sequence models are progressively replacing CNNs as the dominant approach for extracting intricate EEG representations. (3) Evaluating the quality of pre-trained EEG representations before fine-tuning for specific tasks is gradually emerging as a prominent evaluation trend. Besides, compared with the predictive tasks, generative pretext tasks are more challenging and need more trainable parameters and complex structures to accomplish [112], they have advantages in modeling the contextual and temporal dependencies of signals [91].

5 Contrastive-based SSL EEG Analysis Method

Contrastive learning is the most widely used SSL technique in EEG analysis. Contrastive learning frameworks combined with EEG augmentation methods have been investigated to generate representation that integrates invariant features between positive pairs while eliminating irrelevant features between negative pairs. The target of contrastive learning is to encourage the model to pull positive pairs (similar samples) closer together and push negative samples apart in the representation space. Different augmentation methods are applied to EEG signals to form positive and negative pairs. According to the type of augmentation methods for generating positive and negative sample pairs, we can categorize the contrastive-based SSL EEG analysis method into five subcategories: (1) *contrastive predictive coding*, (2) *transformation contrastive learning*, (3) *spatial contrastive learning*, (4) *composite contrastive learning*, and (5) *task-oriented contrastive learning*. The typical frameworks of different kinds of methods are demonstrated in Figure 7 to Figure 11, and the summary of existing works is listed in Table 3.

5.1 Contrastive Predictive Coding

Contrastive Predictive Coding (CPC) is a self-supervised learning technique used in NLP and CV for learning high-level representations [58, 100]. In CPC, data are divided into overlapping

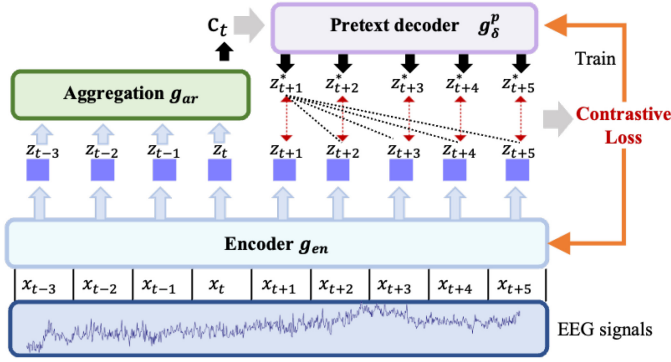


Fig. 7. The framework of CPC for EEG analysis.

context windows, which are used to generate positive and negative pairs. The main idea of CPC is to generate the representation of the context window that can accurately predict the representation of future windows to extract shared invariant features. In the EEG field, two different CPC methods have been investigated:

EEG-based CPC extends the CPC for EEG analysis [12], which can be shown in Figure 7. This method divides EEG signals into time slices through the sliding windows. The context window X_c containing N_c samples is defined as $X_c = \{x_{t_i-N_c+1}, \dots, x_{t_i}\}$, where t_i is the temporal index. Similarly, the following predictive window X_p is defined as $X_p = \{x_{t_i+1}, \dots, x_{t_i+N_p}\}$, where N_p is the length of the prediction window. The encoder g_{en} calculates representation $z_t = g_{en}(x_t)$ for the context window and prediction window, generating context and prediction representation sequences $Z_c = \{z_{t_i-N_c+1}, \dots, z_{t_i}\}$ and $Z_p = \{z_{t_i+1}, \dots, z_{t_i+N_p}\}$, respectively. The integrated feature c_{t_i} is calculated by a GRU-based regression encoder g_{ar} that summarizes the information of representations within the context window. c_{t_i} is used to predict future representations in the prediction window through weight W_k where $k \in [1, N_p]$ and $W_k c_{t_i}$ is the prediction for z_{t_i+k} through the contextual feature c_{t_i} . Positive and negative pairs are then constructed: the predicted representation $W_k c_{t_i}$ forms positive pairs with the corresponding original representation z_{t_i+k} , while forming negative pairs with the remaining representations. The loss function is described as follows:

$$\mathcal{L}_{CPC} = -\frac{1}{|\mathcal{B}|} \sum_{t_i \in \mathcal{B}} \sum_{k=1}^{N_p} \log \frac{\exp(s(x_{t_i+k}, W_k c_{t_i}))}{\exp(s(x_{t_i+k}, W_k c_{t_i})) + \sum_{j \in N_e} \exp(s(x_j, W_k c_{t_i}))}, \quad (14)$$

where $\mathcal{B}$ is the sample batch and $|\mathcal{B}|$ is the batch size. The N_e indexes the negative samples of $W_k c_{t_i}$, where $j \neq k + t_i$. This loss function is a variant of the triplet loss [122], designed to bring positive sample pairs closer together and push negative sample pairs further apart in the representation space. Therefore, the loss function can be simplified as follows:

$$\mathcal{L}_{con} \stackrel{\text{def}}{=} \max(d(x^+, x) - d(x^-, x) - \alpha, 0), \quad (15)$$

where the target of this loss is to train the model to achieve $d(x^+, x) > d(x^-, x) + \alpha$, and α is a small positive number to avoid clustering overfitting. By minimizing the contrastive loss, the model is able to extract invariant temporal features from EEG signals and capture long-term temporal dependencies. This process facilitates the formation of informative representations that retain critical signal information. As a result, the correlation between the learned representations and the raw EEG signals is maximized, which benefits various EEG-based downstream tasks.

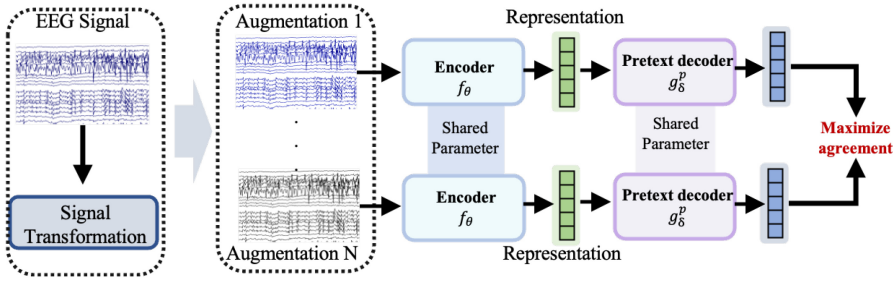


Fig. 8. The framework of transformation contrastive method to capture invariant signal temporal features from unlabeled signals.

EEG-based bidirectional contrastive predictive coding (BCPC) is the extension of CPC to extract bidirectional temporal correlation in EEG signals [22]. Unlike CPC, the BCPC method adds an additional backward prediction window in the framework, representing the EEG signal prior to the context window in the time dimension. The contextual feature c_{t_i} is used to predict the representation in the prediction window and the backward prediction window to construct the positive and negative pairs. By adding the backward prediction window to introduce the reverse negative and positive sample pairs for contrastive learning, the bidirectional model can capture the contextual features with temporal semantic information from both directions in the EEG signal.

5.2 Transformation Contrastive Learning

The transformation contrastive learning method is inspired by the typical contrastive learning framework, such as SimCLR [23] and MoCo [56] in CV. EEG signals are augmented into negative and positive sample pairs through the signal transformation methods designed according to the characteristics of temporal physiological signals. The framework is shown in Figure 8. Multiple transformation contrastive learning methods have been studied to solve different downstream tasks, the typical frameworks are listed as follows:

Signal-transformation contrastive emulates the typical framework SimCLR to conduct EEG contrastive learning [95]. For the randomly selected EEG sample x_t , different transformation methods are employed to generate augmentations $T_1(x_t)$ and $T_2(x_t)$. This method leverages the concept that augmentations applied to the same sample yield similar information, forming positive pairs, while augmentations from distinct samples exhibit significant dissimilarity, constituting negative pairs. The encoder f_θ generates representations, and pretext task decoder g_δ maps the representation into loss space to calculate the contrastive loss. For a batch with size $|B|$, the loss function is defined as follows:

$$\mathcal{L} = -\frac{1}{|B|} \sum_{t=1}^{|B|} \log \frac{\exp(\text{sim}(z_{t_1}, z_{t_2})/\tau)}{\sum_{i=1}^{2k} \mathbb{1}_{[i \neq t]} \exp(\text{sim}(z_{t_1}, z_i)/\tau)}, \quad (16)$$

where z_{t_1} and z_{t_2} are generated by $g_\delta^p(f_\theta(T_1(x_t)))$ and $g_\delta^p(f_\theta(T_2(x_t)))$ respectively, indicating the representations of augmentations from the same sample. $\mathbb{1}_{[i \neq t]} \in \{0, 1\}$ is the indicator function, and τ is the temperature parameter. This loss function represents a variant of the triplet loss, with its simplified form presented in Equation (18). By minimizing this loss function, the model can optimize the representation space to capture discriminative representations. In the EEG analysis domain, EEG augmentation methods can be applied using signal transformation methods mentioned in Section 3.3, and other EEG signal augmentation methods are listed as follows:

(1) *Cutout and resize* divides EEG signals into different segments, and one segment is randomly discarded, representing the “cut out” operation. The remaining segments are then concatenated and resized to the length of the original sample. (2) *Crop and resize* divides EEG signals into different segments, and one segment is randomly chosen and resized to the length of the original sample. (3) *Average filter*, regarded as the smoothing operation, replaces some points in the signal with the value of several neighbor points. (4) *Amplitude scaling*. This method scales the temporal amplitude of the original EEG signal. The scale value should be between 0.5 and 2, as suggested by prior research [95]. (5) *Time shift* method shifts the EEG segments along the time dimension, representing the horizontal offset in temporal sampling. (6) *Direct-current shift* method shifts the EEG segments along the voltage dimension, representing the magnitude offset in temporal sampling. The model can learn invariant EEG features and understand its latent knowledge by conducting contrastive learning on those transformation-augmented EEG signals.

Non-negative EEG contrastive is the contrastive framework without negative samples [152]. In traditional contrastive learning, the quantity and quality of negative samples play a crucial role in determining the effectiveness and quality of contrastive learning. In this framework, z_i and z_j represent the anchor and its positive samples through augmentation. To reduce the impact of negative pairs, this method proposes the *world representation* z_w representing the average information of EEG signal, where $z_w = E_{k \sim p(\cdot)}[z_k]$ is generated by random representation z_k and the distribution $p(\cdot)$. Based on the idea that the similarity between positive pairs should be greater than the similarity between the anchor sample and the global representation, the loss function is designed as follows:

$$l(i, j) = s(z_i, z_w) + \epsilon - s(z_i, z_j), \quad (17)$$

where ϵ is the empirical margin, and $s(\cdot, \cdot)$ is the Gaussian kernel to measure the similarity between input representations. By minimizing the loss, the model makes the similarity between the anchor sample and positive sample greater than the world representation to learn consistent EEG information between samples without human labels. Besides, some EEG analysis methods are integrated with non-negative contrastive frameworks in CV like Barlow Twins [158] and BYOL [49] to conduct EEG-based non-negative contrastive learning, where all the augmented samples form the positive pairs with the anchor sample and the well-designed loss function can extract invariant features from only positive pairs. Those methods are also non-negative contrastive frameworks without global representation.

5.3 Spatial Contrastive Learning

The spatial contrastive learning method shown in Figure 9 focuses on spatial information and utilizes channel-level spatial augmentation techniques (e.g., jigsaw, meiosis) on EEG signals to construct positive and negative sample pairs, from which the model can integrate efficient spatial features and channel correlation into representation. Typical methods are listed as follows:

Spatial shuffle contrastive method conducts the channel-shuffling technique to construct positive and negative pairs [86]. In this method, EEG signal $x \in \mathbb{R}^{c \times t}$ is augmented through spatial shuffle: different EEG channels are categorized into different brain regions based on their spatial positions, generating $X^B = \{X^1, X^2, \dots, X^M\}$, where M is the number of brain regions and each region in X^B contains features from multiple channels. X^B is randomly shuffled and reassembled into the augmented EEG sample $X^* \in \mathbb{R}^{c \times t}$. Each sample generates two shuffling augmentations to form positive sample pairs, and shuffling augmentations from different samples form negative pairs. InfoNCE loss described in Equation (19) serves as the loss function for model training. The model can understand relationships between spatial channel location and signal features by contrasting the shuffling augmented EEG samples.

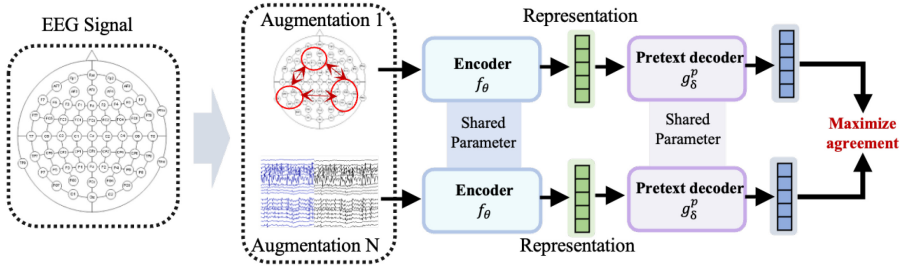


Fig. 9. The framework of spatial contrastive method. In this framework, various channel-level spatial augmentation methods (e.g., channel shuffle, channel meiosis) are used to construct positive and negative sample pairs for contrastive learning, where the model can capture invariant spatial features and channel correlations from unlabeled EEG samples.

Graph contrastive method mines the relationship between channels using the graph structure [61, 154]. In this framework, EEG signals are embedded into node features in the graph, and the edges between nodes are calculated by the channel correlation or the spatial distance. Assuming $\mathcal{G}$ is the generated graph, $\mathcal{V}$ is the node set, and $\mathcal{E}$ is the edge set, two augmentation methods are employed for contrastive learning: (1) *node dropping method*. For a sample $\mathcal{G}_t$, two augmented samples $\mathcal{G}_t^1$ and $\mathcal{G}_t^2$ are generated by randomly dropping nodes and their edges according to the dropping rate $r\%$. The augmentations from the same sample form the positive pairs, and from different samples form the negative pairs. (2) *Subgraph augmentation*. For each node v_i in the sample $\mathcal{G}_t$, two positive and one negative samples are constructed: the random walk with restart algorithm is used to generate positive subgraphs $\mathcal{G}_{i,1}^+$ and $\mathcal{G}_{i,2}^+$ centered at the selected node v_i with the radius parameter ra to control the size, and generates negative subgraph $\mathcal{G}_i^-$ centered at the farthest node from the selected node. In the positive subgraph, the features of target nodes are masked with zero to avoid interference from target node information. For different sub-graphs, the representations are encoded by the trainable weight W_e through the GNN, where $re_{i,1}^+$ and $re_{i,2}^+$ are the representations for positive samples, and $re_{i,1}^-$ is the representation of the negative sample. The embedding of the selected node is also calculated by W_e , where $e_i = \text{ReLU}(v_i W_e)$. A trainable score matrix W_s is then designed to quantify the similarity between the selected node and its subgraphs, which is described as follows:

$$S_{i,j}^+ = \sigma \left(e_i W_s re_{i,j}^+ \right), \quad (18)$$

where σ represents the logistic function. The contrastive loss is designed to maximize the correlation between the embedding of nodes and positive samples, which makes the representation of channels in latent space closer to similar channels. The loss function is defined as

$$\mathcal{L} = -\frac{1}{2c|\mathcal{B}|} \sum_{j=1}^2 \sum_{i=1}^c \left(\log \left(S_{i,j}^+ + \log(1 - S_{i,j}^-) \right) \right), \quad (19)$$

where c is the number of channels and $|\mathcal{B}|$ is the batch size. By minimizing this loss, the model focuses on channel-level spatial features, which confers the model with the robust ability to comprehend high spatial resolution in EEG, leading to superior performance in downstream tasks involving multiple channels and complex channel configurations.

EEG meiosis contrastive method conducts meiosis augmentation technique and contrastive learning framework to integrate invariant channel features into representation [51]. The meiosis data augmentation technique is used to generate contrastive pairs: two EEG samples are randomly

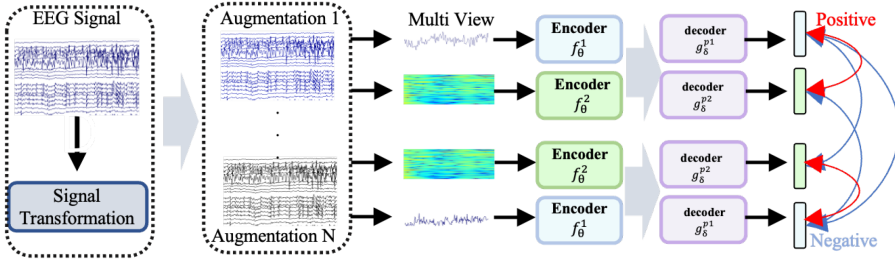


Fig. 10. An example of the composite contrastive learning framework (temporal-frequency contrastive method).

sampled into the group $X_i^g = \{A_i, B_i\}$, where the format of samples A_i and B_i are $\mathbb{R}^{c \times t}$, c is the channel number and t is the number of sampling points. For the group, $A_i = \{a_1, a_2, a_3, \dots, a_t\}$ and $B_i = \{b_1, b_2, b_3, \dots, b_t\}$ are engaged into meiosis with each other, which signifies data exchange between A^p and B^p , generating the augmented sample $V_i^1 = \{a_1, a_2, a_3, \dots, a_i, b_{i+1}, b_{i+2}, \dots, b_c\}$ and $V_i^2 = \{b_1, b_2, b_3, \dots, b_i, a_{i+1}, a_{i+2}, \dots, a_c\}$. EEG samples A_i and B_i are under the same stimulus/event to increase contrasting complexity. All training samples are augmented and transformed into V_i^1 and V_i^2 for contrast, and the sample feature representation can be generated through encoder f_θ and projector (pretext task decoder) g_θ^p by $z_i^1 = g_\theta^p(f_\theta(V_i^1))$. In the framework, z_i^1 and z_i^2 form the positive pair, indicating samples exchanged EEG signal with each other, while z_i^1 and z_j^2 form the negative pair ($i \neq j$). The loss function is then defined as follows, which is the variant of triplet loss:

$$L = -\frac{1}{2} \left(\frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \log \frac{\exp(s(z_i^1, z_i^2)/\tau)}{\sum_{j=1}^{|\mathcal{B}|} \mathbb{1}_{[j \neq i]}(s(z_i^1, z_j^1)/\tau) + \sum_{j=0}^{|\mathcal{B}|} (s(z_i^1, z_j^2)/\tau)} + \frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \log \frac{\exp(s(z_i^1, z_i^2)/\tau)}{\sum_{j=1}^{|\mathcal{B}|} \mathbb{1}_{[j \neq i]}(s(z_i^2, z_j^2)/\tau) + \sum_{j=0}^{|\mathcal{B}|} (s(z_i^1, z_j^2)/\tau)} \right), \quad (20)$$

where $s(\cdot, \cdot)$ is the function to measure the similarity between representations, and $\mathbb{1}_{[j \neq i]} \in \{0, 1\}$ is the indicator function that equals 0 when $i = j$. The proposed contrastive loss aims at minimizing the distance between mutually coupled sample pairs (V_i^1, V_i^2), and maximize the distance between other sample pairs without mutual coupling: (V_i^1, V_j^1), (V_i^2, V_j^2), and (V_i^1, V_j^2), where $i \neq j$. By minimizing the loss function, the model is trained to comprehend specific and coherent channel features and can discriminate homologous EEG channel data, which can be regarded as the model capturing the EEG channel distribution knowledge, proving highly beneficial for EEG-based tasks.

5.4 Composite Contrastive Learning

Composite contrastive learning is a complex framework that enhances EEG signals through multi-view or multi-domain augmentations. It performs cross-view and cross-domain contrastive learning to extract more expressive and complex representations that integrate domain-specific signal knowledge. Figure 10 shows an example of the typical framework, and existing composite EEG contrastive learning frameworks are listed as follows:

Frequency-temporal contrastive method conducts contrastive learning on temporal and frequency domain. Two different frequency-temporal contrastive strategies have been investigated:

(1) *Complementary strategy* conducts cross-view contrastive learning to avoid the ignorance of complementary information in different views [79]. EEG signal x_i is augmented into $x_{i,1}$ and $x_{i,2}$

through signal augmentations, which are then mapped into the temporal and spectral domains independently, generating temporal components $x_{i,1}^t, x_{i,2}^t$ and spectrum components $x_{i,1}^s, x_{i,2}^s$. Different augmentations are subsequently processed through the temporal encoder f_θ^t and spectrum encoder f_θ^s to construct the representations $z_{i,1}^s$ and $z_{i,1}^t$ from augmentation $x_{i,1}$, and $z_{i,2}^s$ and $z_{i,2}^t$ from augmentation $x_{i,2}$. Four losses are combined to train the model: (1) Temporal contrastive loss, denoted as $\mathcal{L}_{tt}$. The temporal representations generated from same augmentation form positive pairs $\{z_{i,1}^t, z_{i,2}^t\}$, and from different augmentation form negative pairs $\{z_{i,1}^t, z_{i,2}^t\}, i \neq j$, with the InfoNCE serves as the loss function. (2) Spectrum contrastive loss, denoted as $\mathcal{L}_{ss}$ calculated by spectral augmented representations similar to temporal contrastive loss. (3) Mixing contrastive loss, denoted as $\mathcal{L}_{gg}$. The spectrum and temporal augmented representations are concatenated to form the mixing augmented representation, where $z_{i,1}^g = \text{cat}(z_{i,1}^t, z_{i,1}^s)$, cat represents the concatenation operation. This loss can be calculated similarly to the first two losses. (4) Complementary loss, denoted as $\mathcal{L}_d$. The above losses may narrow the distance between representations, losing complementary features in each view. Therefore, the complementary loss is designed to pull corresponding augmented samples in the same view closer while pushing away the corresponding augmented samples in different views. Assuming $z_i = \{z_{i,1}^t, z_{i,2}^t, z_{i,1}^s, z_{i,2}^s\}$, complementary loss is defined as

$$\begin{cases} l_d(z_i, j, k) = -\log \frac{\exp(s(z_i[j], z_i[k])/\tau)}{\sum_{q=1}^4 \mathbb{1}_{[q \neq j]} \exp(s(z_i[j], z_i[q])/\tau)} \\ \mathcal{L}_D = \frac{1}{4|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} l_d(z_i, 1, 2) + l_d(z_i, 2, 1) + l_d(z_i, 3, 4) + l_d(z_i, 4, 3) \end{cases}, \quad (21)$$

where $s(\cdot, \cdot)$ is the similarity function. Multiple loss functions are combined to train the model to extract multi-domain features and preserve the domain-specific and complementary features, which can be described as follows:

$$\mathcal{L}_{con} = \lambda_1(\mathcal{L}_{tt} + \mathcal{L}_{ss} + \mathcal{L}_{gg}) + \lambda_2 \mathcal{L}_D, \quad (22)$$

where λ_1 and λ_2 are the hyperparameters to balance different contrastive losses.

(2) *Consistent strategy* focuses on extracting consistent information between temporal and frequency representations through contrastive learning [167]. Different from the complementary strategy, this strategy aims at maximizing the mutual information between temporal and frequency representations to align different representations in a latent feature space to extract multi-domain coherent features. The consistent loss function is described as follows:

$$\mathcal{L}_c = \frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \sum_{Sim^*} (Sim^{t_1, f_1} - Sim^* + \delta), Sim^* \in \{Sim^{t_1, f_2}, Sim^{t_2, f_1}, Sim^{t_2, f_2}\}, \quad (23)$$

where δ is the hyperparameter. In this loss function, the $Sim_i^{t_1, f_1} = d(z_{i,1}^t, z_{i,1}^f)$ is defined to measure the representation similarity, where $z_{i,1}^t$ and $z_{i,1}^f$ are the temporal and frequency representations generated by sample x_i , and $z_{i,2}^t$ and $z_{i,2}^f$ are generated by augmented sample x_i^* . By minimizing this loss, the frequency and temporal representations can be pulled closer for a sample in the latent space to mine for multi-domain consistent features.

Multi-view CPC extends CPC from the single view to multiple views for exploring complex EEG features [39]. In this method, the weak and strong augmentation methods are designed to construct two views: jitter-and-scale strategy is used to construct weak augmentations x_i^1 , while permutation-and-jitter generates complex strong augmentation x_i^2 . According to the definition of

CPC, the integrated features c_i^1 and c_i^2 of context windows from two views are generated. Cross-view prediction strategy is implemented, where c_i^1 is used to predict future windows z_{i+1}^2 and c_i^2 is used to predict future windows z_{i+1}^1 , generating CPC losses $\mathcal{L}_{1,2}$ and $\mathcal{L}_{2,1}$. Besides, the cross-view contextual contrastive strategy is designed to extract discriminative features: c_i^1 and c_i^2 generated from the same sample but in different views form the positive pairs, while other representations form negative pairs. For the samples in batch $\mathcal{B}$, one given sample can construct one positive pair and $2|\mathcal{B}| - 2$ negative pairs. The loss function is defined as follows, which is the variant of triplet loss:

$$\mathcal{L}_{CC} = - \sum_{i=1}^{|\mathcal{B}|} \frac{\exp(s(c_i^1, c_i^2)/\tau)}{\sum_{j=1}^{|\mathcal{B}|} \sum_{q=1}^2 \mathbb{1}_{[j \neq i]} \exp(s(c_i^1, c_j^q)/\tau)}. \quad (24)$$

Different loss functions are combined as $\mathcal{L} = \lambda_1(\mathcal{L}_{1,2} + \mathcal{L}_{2,1}) + \lambda_2 \mathcal{L}_{CC}$ through the weight λ_1 and λ_2 to balance different losses. The multi-view CPC method can mine complex temporal features and understand the aligned representation.

Multi-level contrastive method conducts contrastive learning at multiple levels to capture complex signal features [161]. In this framework, EEG sample is divided into n_l segments: $X_i = \{x_1, x_2, x_3, \dots, x_{n_l}\}$, with CNN encoder f_θ^1 to generate local representation $Z_i = \{z_1, z_2, z_3, \dots, z_{n_l}\}$ and transformer encoder f_θ^2 to generate contextual representation $R_i = \{r_1, r_{1+k}, r_{1+2k} \dots\}$, where contextual representation r_j is generated by the integration of k neighbor local representations. The positive and negative sample pairs are constructed according to the filter with EEG rhythm rules. InfoNCE loss is used at local and contextual levels to extract multi-granular features. The fusion of multi-level contrastive learning can integrate different aspects of temporal features of EEG signals into representation, making representation more efficient and expressive.

Scalp-dipole neural contrastive is a knowledge-based cross-view contrastive method to generate general neural representation [143]. In this framework, two views are constructed according to the neural source [65] of EEG signal x_i : the scalp view sc_i is constructed by spatial matrix, indicating the distribution of EEG voltage across the scalp; the dipole view dp_i is constructed by undirected graph, indicating the inner correlation of dipoles (activated pyramidal cells) that produce the EEG signals. EEG signals are augmented through mask and jigsaw, and then also transformed into two views. The CNN encoder f_θ^c and graph convolutional encoder f_θ^g are designed to generate scalp and dipole representation from different views, and two contrastive strategies are proposed: (1) *Inner-view contrastive* aims at extracting invariant features in each view. Augmented samples in a specific view are considered as positive pairs and the Barlow Twins loss [158] is implemented to minimize their distance in the representation space and capture invariant information between augmented samples. (2) *Cross-view contrastive* is based on the theory that EEG representations for different views are homogenous and contain similar neural information. Therefore, augmentations in two views generated by the same sample construct positive pairs and augmentations in two views construct negative pairs, with InfoNCE loss to train the model. Combining the inner-view and cross-view losses can extract both view-specific features and latent neural information, generating general representations effective for different EEG-based tasks.

5.5 Task-oriented EEG Contrastive Learning

Task-oriented contrastive learning is an idiosyncratic framework set up to solve specific tasks. Some specific contrastive frameworks are designed for particular tasks: (1) **Image-EEG contrastive** [129], where the image and the corresponding EEG signals elicited by viewing this image can form positive pairs, and the image with other EEG signals can form negative pairs. The process of image-EEG contrastive learning can be shown in Figure 11. The mutual information between the

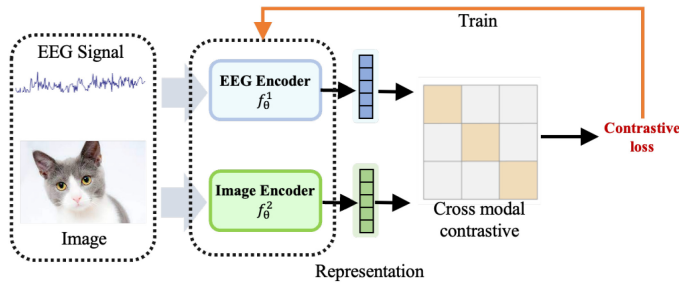


Fig. 11. An example of task-oriented EEG contrastive learning: where the contrastive sample pairs are constructed from EEG signal and image data to align the features of EEG signal with image feature.

image and its corresponding EEG signal is maximized by image-EEG contrastive, which can solve the task of *EEG image decoding*. (2) **Speech-EEG contrastive learning** is inspired by the **contrastive language-image pre-training (CLIP)** [111] method that forms the EEG-speech sample pairs to extract correlation and solve the *EEG speech decoding* task [32]. (3) **Cross-subject contrastive learning** aims at addressing the individual variability issue in EEG signals. For example, *age contrastive learning* selects anchor samples from different age groups. Samples with a small age difference from the anchor are treated as positive samples to form positive pairs, while those with a large age difference are considered negative samples to form negative pairs. By contrasting negative and positive pairs, the model can capture the age-related brain features to improve the generalizability of generated representations [137].

5.6 Section Discussion

This section comprehensively reviews various contrastive-based EEG analysis methods. We categorize the contrastive-based frameworks into five subcategories: (1) Contrastive predictive coding method that integrates the prediction and contrastive tasks to capture temporal information. (2) Transformation contrastive learning to extract signal-related invariant features. (3) Spatial contrastive method to capture spatial channel correlation. (4) Composite contrastive method that conducts multi-view contrastive learning to extract spatial-temporal-spectral features. (5) Task-oriented contrastive method that constructs specialized framework toward specific tasks. The contrastive-based SSL EEG analysis method is summarized in Table 3, highlighting significant advancements in the field. The transformation-based contrastive tasks are the most widely adopted, representing a direct adaptation of contrastive learning techniques from computer vision (e.g., SimCLR [23], MOCO [56]) to EEG analysis. The Transformer-based contrastive tasks have proven instrumental in downstream tasks such as sleep stage classification, seizure detection, and pathology detection, demonstrating their ability to effectively extract invariant patterns and critical information from augmented EEG signal samples. The contrastive predictive coding and spatial contrastive pretext tasks have also been widely studied and applied in various EEG downstream tasks, highlighting the importance of the temporal dependency features and spatial (channel-wise) invariant features of EEG signals. Additionally, composite contrastive tasks focus on the time-frequency characteristics of EEG signals, exploring the relationships and invariant information across the time and frequency domains. These pretext tasks are specifically tailored for EEG-based downstream tasks that heavily rely on frequency spectral features, such as sleep stage classification and seizure detection. Regarding architectures, CNNs remain the dominant backbone, frequently combined with recurrent models like LSTMs (e.g., SleepDPC) or GRUs (e.g., Seq-SimCLR) to effectively model spatial and temporal features. Emerging models

Table 3. The Summarization of Contrastive-based EEG Analysis SSL Method

Approach	Sub-category	Detailed Method	Backbone	Downstream Tasks	Training Mode
Clinical EEG SSL [12]	Contrastive predictive coding	EEG-based CPC	CNN	Sleep and pathology classification	PT
SSL for EEG [11]	Contrastive predictive coding	EEG-based CPC	CNN	Sleep classification	PT
ContrastWR[152]	Transformation-based contrastive	Non-negative CL	CNN	Sleep and pathology classification	LE
SSCL for EEG[68]	Transformation-based contrastive	Signal transformation CL	CNN	Sleep classification	PT&LE
EEG-CGS[61]	Spatial contrastive	Spatial shuffle CL	GNN	Seizure analysis	PT
GMSS [86]	Spatial contrastive	Graph-based CL	GNN	Emotion recognition	PT&LE&JT
SleepDPC [148]	Contrastive predictive coding	EEG-based CPC	CNN&LSTM	Sleep classification	LE
Seq-SimCLR [95]	Transformation-based contrastive	Signal transformation CL	CNN&GRU	Multiple tasks	PT&LE
Domain-guide CL[137]	Task-oriented contrastive	Cross-subject CL	CNN	Multiple tasks	PT
Multivariate CL[16]	Spatial contrastive	Graph-based CL	GNN	Sleep classification	PT
DS-AGC[154]	Spatial contrastive	Graph-based CL	GNN	Emotion recognition	PT
ME-MHAC[51]	Spatial contrastive	Meiosis-based CL	CNN	Emotion recognition	PT
MBrain[17]	Contrastive predictive coding	EEG-based CPC	CNN&LSTM	Seizure detection	PT
BrainNet[22]	Contrastive predictive coding	Bidirectional CPC	GNN	Seizure detection	JT
SleepECL[161]	Composite contrastive	Multi-level CL	Transformer	Sleep classification	LE
TS-TCC[39]	Composite contrastive	Multi-view CPC	Transformer	Sleep and seizure detection	PT&LE
CoSleep[153]	Contrastive predictive coding	EEG-based CPC	CNN	Sleep classification	LE
SLAM-EEG[149]	Transformation-based contrastive	Transformation-based CL	ViT	Seizure detection	PT
SPP-EEGNET[85]	Transformation-based contrastive	Transformation-based CL	CNN	Multiple tasks	PT
DSSNet[19]	Contrastive predictive coding	EEG-based CPC	CNN&RNN	Sleep classification	LE
TF-C[167]	Composite contrastive	Frequency-temporal CL	CNN&Tfm	Sleep classification	LE
TS-MoCo[52]	Transformation-based contrastive	Transformation-based CL	Transformer	Emotion recognition	PT&LE
MV-EEG[62]	Composite contrastive	Frequency-temporal CL	Transformer	Pathology detection	LE
PSN-Sleep[155]	Transformation-based contrastive	Non-negative CL	CNN	Sleep classification	LE
MulEEG[79]	Composite contrastive	Frequency-temporal CL	CNN	Sleep classification	LE
MI-SSLEEG[53]	Transformation-based contrastive	Transformation-based CL	CNN	Motor imagery	JT
SA-EEG[26]	Transformation-based contrastive	Transformation-based CL	CNN	Motor imagery	LE
MtCLSS[138]	Transformation-based contrastive	Transformation-based CL	CNN	Sleep classification	LE
Multi-channel CL[43]	Transformation-based contrastive	Transformation-based CL	CNN	Sleep and pathology classification	LE
SGMC [72]	Spatial contrastive	Meiosis-based CL	CNN	Emotion recognition	PT
CLISA [123]	Task-oriented contrastive	Cross-subject CL	CNN	Emotion recognition	LE
KDC[143]	Composite contrastive	Scalp-dipole neural CL	CNN&GNN	Multiple tasks	PT&LE
NICE-EEG[129]	Task-oriented contrastive	Image-signal CL	ViT&GNN	Image-decoding	LE
AAD[32]	Task-oriented contrastive	Speech-signal CL	CNN&LSTM	Speech-decoding	LE

“CPC” represents contrastive predictive coding, “CL” represents contrastive learning, “PT” represents pre-training and fine-tune mode, “LE” represents linear evaluation mode, and “JT” represents joint-training mode.

increasingly leverage Transformers (e.g., TS-TCC) or GNNs (e.g., GMSS, BrainNet) to handle complex graph-based or sequence modeling tasks, indicating a shift toward advanced architectures in EEG analysis. The training modes show a preference for pre-training and fine-tuning, while linear evaluation is used to assess the quality of learned EEG representations. Compared to other EEG SSL frameworks, contrastive-based tasks are less computationally intensive, with fewer parameters and simpler tasks to generate effective representations with higher generation and information density (also supported by [112]). Contrastive methods rely on augmentation techniques, where the well-designed sample pairs can help the model integrate critical neural knowledge, and arbitrarily chosen sample pairs may yield counterproductive results.

6 Hybrid SSL EEG Analysis Method

The hybrid SSL EEG analysis method combines various pretext tasks to jointly train the model to learn complex knowledge or information. The idea of multi-task learning [75, 170] has been applied in hybrid SSL methods: the common encoder f_θ is used to extract features and generate

Table 4. The Summarization of Hybrid-based EEG Analysis Self-supervised Learning Method

Approach	Pretext-category	Backbone	Downstream Tasks	Training Mode
Clinical EEG SSL [12]	Predictive task/contrastive task	CNN	Sleep and pathology classification	PT
SSL for sleep EEG [11]	Predictive task/contrastive task	CNN	Sleep and pathology classification	PT
EEG-CGS [61]	Generative task/contrastive task	CNN	Seizure analysis	PT
GMSS [86]	Predictive task/contrastive task	GNN	Emotion recognition	PT&LE
MBrain [17]	Predictive task/contrastive task	CNN&LSTM	Seizure detection	PT
MtCLSS [87]	Predictive task/contrastive task	CNN	Sleep classification	LE

“PT” represents pre-training and fine-tune mode, “LE” represents linear evaluation mode.

representation from EEG signal, with different pretext task decoders $\{g_{\delta}^{p_1}, g_{\delta}^{p_2}, \dots\}$ used to solve multiple pretext tasks. The losses from different tasks are fused to train the model, where the shared encoder can fully leverage the advantages of different tasks to obtain representation that encompasses more knowledge and exhibits stronger expressive capabilities. The combination of multi-task losses with weight λ can be described as follows:

$$\mathcal{L}_{mt} = \sum_{i=1}^{t_n} \lambda_i \mathcal{L}_i. \quad (25)$$

Table 4 shows the existing hybrid EEG SSL methods. In the existing studies, different combinations of pretext tasks are used to generate representations: many methods combine the predictive and contrastive tasks [11, 12], where the decoders predict the augmentations and conduct negative and positive pairs for contrastive learning to capture critical discriminative information and invariant features; another method combines the generative and contrastive tasks [61] to explore the local correlations and global coherent features of EEG signals. Although the hybrid SSL method can capture complex features through multiple tasks, the gradient interference caused by training various tasks may influence the effectiveness of the generated representation. Therefore, this method requires careful selection of correlated pretext tasks to avoid interference between tasks.

7 Future Directions

In the EEG analysis field, combining deep models and SSL frameworks can help improve the model performance on various EEG-based tasks through extra parameter training on unlabeled EEG samples with well-designed pretext tasks. In addition to the advantages of EEG-based SSL frameworks, we analyze the challenges in the existing EEG-based SSL studies and propose potential future directions for EEG-based SSL to address the challenges and problems.

Signal-oriented pretext task. Most existing pretext tasks are the straightforward extension of pretext tasks in CV and NLP, which treat EEG signals as 2D matrix and temporal vector like image or text patches to capture spatial and contextual correlations but ignore the intrinsic characteristics of EEG signal. Therefore, designing the EEG-oriented pretext task to extract the spatial-temporal-frequency EEG features is a feasible approach worth further exploration.

Knowledge-driven SSL framework. Although SSL frameworks have achieved significant success in various EEG-based tasks, the lack of theoretical foundation and neural knowledge for EEG signals leads to the generated representations lacking generalization and interpretability. Therefore, how to integrate the EEG-based neural knowledge with the SSL framework to construct the knowledge-driven interpretable EEG model is another important direction, which needs to design specific pretext tasks and augmentation techniques that can fuse explainable neural knowledge into representation. We believe that by integrating knowledge of EEG into the

self-supervised framework, the models are expected to bring generalization and interpretability to representations.

Graph-based SSL. Deep learning models like CNN and RNN have been widely used to extract spatial-temporal features from EEG signals for different tasks. However, most existing methods ignore the inherent topological connections among electrodes. EEG signals are generated from the activity of neurons that are topologically connected inner the brain. Graph neural networks can explore the inherent connectivity patterns among neurons, and we believe that researching GNN-based EEG SSL methods can integrate richer latent brain information into representations, offering a new perspective for information expression.

SSL for Heterogeneous EEG. The ultimate goal of SSL for EEG analysis is to generate general representations for various downstream tasks. However, EEG signals are collected from multiple scenarios which encompass variations in channel, device, sampling rate, task, subject, and distribution. The significant differences between EEG signals from different sources make it challenging for self-supervised training collaboratively. Therefore, constructing SSL framework tailored for heterogeneous EEG data is an important direction for future development. Exploration in this direction can leverage heterogeneous EEG samples from multiple sources to jointly pre-train the model. Such an approach enables the effective use of diverse EEG datasets to learn universal representations that generalize well across different downstream tasks.

Multi-modal SSL. SSL for EEG signals is the unimodal approach aiming at extracting neural information from unlabeled EEG samples. However, the features mined from EEG signals are difficult to adapt to some complex downstream tasks, which require other brain or physiological signals to provide more abundant information. Therefore, the EEG-based multimodal self-supervised learning method needs to be further studied to extract integrated and aligned features from unlabeled multimodal signals (ECG, EMG, EOG, etc.) for challenging downstream tasks.

8 Conclusion

This article is a comprehensive survey of self-supervised learning for EEG analysis, including the reasonable taxonomy, different kinds of existing EEG-based SSL methods, downstream EEG tasks, and the available training datasets, offering detailed guidelines for researchers interested in deep learning combined with EEG analysis. We first review typical SSL frameworks and pretext tasks in the CV and NLP and introduce traditional supervised EEG analysis methods as the preliminary, to illustrate the drawbacks of supervised EEG analysis and underscore the necessity of introducing SSL for EEG analysis. We then provide a detailed exposition on four categories of SSL frameworks for EEG analysis, elucidating the technical details of representative methods to extract spatial-temporal-frequency features from EEG signals. Subsequently, we enumerate EEG-based downstream tasks effective for SSL frameworks and present relevant EEG datasets suitable for pre-training or downstream task fine-tuning. Finally, we discuss the challenges in existing studies and propose new insights and potential future directions that are worth exploring, which can help generate more general and explainable representations to address various complex downstream tasks.

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